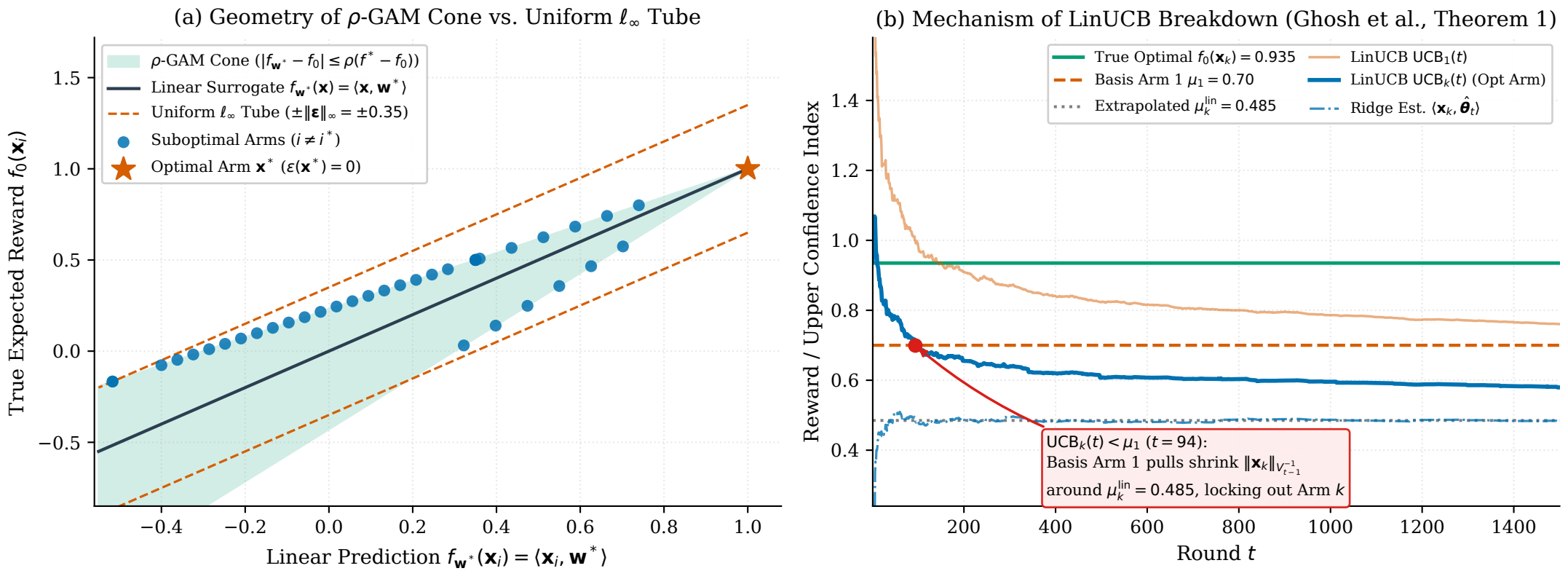

(a) Geometry of ρ-GAM Cone vs. Uniform ℓ_∞ Tube
ρ-GAM Cone ($|f_{\mathbf{w}^*} - f_0| \le \rho(f^* - f_0)$)
Linear Surrogate $f_{\mathbf{w}^*}(\mathbf{x}) = \langle \mathbf{x}, \mathbf{w}^* \rangle$
Uniform ℓ_∞ Tube ($\pm\|\boldsymbol{\varepsilon}\|_\infty = \pm 0.35$)
Suboptimal Arms ($i \neq i^*$)
Optimal Arm $\mathbf{x}^*$ ($\varepsilon(\mathbf{x}^*) = 0$)
True Expected Reward $f_0(\mathbf{x}_i)$
Linear Prediction $f_{\mathbf{w}^*}(\mathbf{x}_i) = \langle \mathbf{x}_i, \mathbf{w}^* \rangle$
(b) Mechanism of LinUCB Breakdown (Ghosh et al., Theorem 1)
True Optimal $f_0(\mathbf{x}_k) = 0.935$
Basis Arm 1 $\mu_1 = 0.70$
Extrapolated $\mu_k^{\text{lin}} = 0.485$
LinUCB $\text{UCB}_1(t)$
LinUCB $\text{UCB}_k(t)$ (Opt Arm)
Ridge Est. $\langle \mathbf{x}_k, \hat{\boldsymbol{\theta}}_t \rangle$
$\text{UCB}_k(t) < \mu_1$ ($t = 94$):
Basis Arm 1 pulls shrink $\|\mathbf{x}_k\|_{V_{t-1}^{-1}}$
around $\mu_k^{\text{lin}} = 0.485$, locking out Arm k
Reward / Upper Confidence Index
Round t